Background: Miscellaneous, History and References

Sections 2.8 – 2.9

Chapter 2 — Background

An Introduction to Universal Artificial Intelligence

Hutter, Quarel, Catt (2024)

Outline

What This Section Is For

Section 2.8 collects material that quietly underlies the rest of the book, but which is more of a technical toolbox than essential reading on a first pass. It is organized into three parts.

2.8.1 Distances and their relation

Several different notions of "distance" between two probability distributions (absolute, squared, KL, Hellinger) are introduced, and we prove that the Kullback–Leibler (KL) divergence upper-bounds all the others. This lets us prove one bound and get several bounds for free.

2.8.2 Dealing with semimeasures

A *semimeasure* is a relaxation of a probability measure that allows total probability to be less than 1 (some mass can be "missing"). This subsection lists workarounds for the extra technical headaches semimeasures cause.

2.8.3 Probability zero

How to sensibly define conditional probabilities and expectations when we condition on an event that has probability exactly zero.

Why We Need Multiple Notions of Distance

Throughout the book (and especially in Chapter 3) we need to measure how "far apart" two probability distributions P and Q are. Several different distances are useful, each with its own pros and cons:

- **Absolute distance** (twice the total variation distance): a proper metric, very natural for probability measures.
- **Squared distance** (square of the Euclidean distance): the most common choice, but often the least suitable one (see the exercises).
- **KL divergence** (Section 2.5.3): not a metric (it is not symmetric and can be infinite), but it is the information-theoretically most useful quantity for proving bounds.
- **(Squared) Hellinger distance**: another proper metric, useful in some later chapters.

Key idea of this subsection. KL divergence is an upper bound to all the other distances, while at the same time being a lower bound for many loss functions used later in the book. This makes it an extremely useful hub through which to route proofs. We restrict attention to finite sample spaces Ω, where probabilities are d-dimensional vectors in the probability simplex.

Lemma 2.8.1: Pinsker's Binary Inequality

Lemma 2.8.1 (Pinsker's binary inequality)

For $0 \leq p, q \leq 1$,

$$2(p-q)^2 \;\leq\; p \ln \frac{p}{q} + (1-p) \ln \frac{1-p}{1-q}.$$

Plain English. Think of $p = P(1)$ and $q = Q(1)$ as the probabilities that two coins (biased distributions P and Q over $\{0, 1\}$) land on "1". The right-hand side is exactly the KL divergence between P and Q. The lemma says: twice the squared gap between the two coin biases is never bigger than their KL divergence. In other words, KL divergence controls (upper-bounds) the simpler squared distance.

There are many known proofs of this inequality, none of them particularly illuminating; we present the shortest one.

Proof of Pinsker's Binary Inequality I

Proof. Define $f(x) = p \ln x + (1-p)\ln(1-x)$. Then

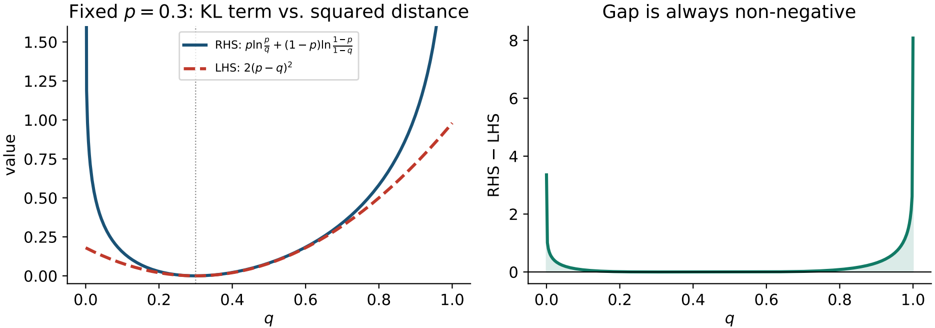
$$p \ln \frac{p}{q} + (1-p)\ln\frac{1-p}{1-q} \;=\; f(p) - f(q).$$

Plain English. f is just a convenient helper function built from p; subtracting f at two points p and q reproduces exactly the KL term we want to bound.
Also note that for $0 \le x \le 1$ we have $x(1-x) \le 1/4$ (the maximum of $x(1-x)$ on $[0,1]$ is $1/4$, attained at $x = 1/2$). Then

$$f(p) - f(q) \;=\; \int_q^p f'(x)\,dx \;=\; \int_q^p \frac{p-x}{x(1-x)}\,dx \;\ge\; 4\int_q^p (p-x)\,dx \;=\; 2(p-q)^2. \qquad \blacksquare$$

Plain English. We rewrite the difference $f(p) - f(q)$ as an integral of the derivative f', then use $x(1-x) \le 1/4$ (so $1/(x(1-x)) \ge 4$) to lower-bound the integrand, and finally evaluate the resulting simple integral to get exactly $2(p-q)^2$.

Proof of Pinsker's Binary Inequality II



Left: both sides of the inequality as functions of q, for fixed $p = 0.3$. Right: their gap, always non-negative.

Lemma 2.8.2: Convex Even Functions Are Monotonic on $\mathbb{R}^+$

Lemma 2.8.2 (Convex even functions are monotonic on $\mathbb{R}^+$)

Let $f : \mathbb{R} \to \mathbb{R}$ be an *even* convex function (i.e. $f(x) = f(-x)$ for all x). Then f is monotonically increasing over the positive reals.

Plain English. "Convex" means the graph of f curves upward (a chord between two points on the graph lies above the graph); "even" means the graph is symmetric about the vertical axis, like $f(x) = x^2$ or $f(x) = |x|$. The lemma says any such function, restricted to positive inputs, can only go up (or stay flat) as its input increases — it is a purely geometric fact used repeatedly in the next few results.

Proof. Take the definition of convexity, $f(\alpha x + (1-\alpha)y) \leq \alpha f(x) + (1-\alpha)f(y)$ for $0 \leq \alpha \leq 1$, and insert $x = a+b$, $y = 0$:

$$f(\alpha(a+b)) \;\leq\; \alpha f(a+b) + (1-\alpha)f(0). \tag{2.8.3}$$

Plain English. α (alpha) is a mixing weight between 0 and 1; this is just the convexity inequality applied to the two specific points $a+b$ and 0.
Now, assuming $a, b \geq 0$, insert $\alpha = \frac{a}{a+b} \geq 0$ or $\alpha = \frac{b}{a+b} \geq 0$ to get two inequalities:

$$f(a) \;\leq\; \frac{a}{a+b}f(a+b) + \frac{b}{a+b}f(0), \qquad f(b) \;\leq\; \frac{b}{a+b}f(a+b) + \frac{a}{a+b}f(0).$$

Adding the two:

$$f(a) + f(b) \;\leq\; f(a+b) + f(0). \tag{2.8.4}$$

Proof of Lemma 2.8.2 II

Also, in (2.8.3) choose $\alpha = 1/2$, $x = -b$, $y = b$, to obtain

$$f(0) \;\le\; \tfrac{1}{2}f(b) + \tfrac{1}{2}f(-b) \;=\; f(b)$$

since f is even (so $f(-b) = f(b)$).
Plain English. Plugging in a mix of $-b$ and b (which averages to 0) into the convexity inequality tells us $f(0)$ can never exceed $f(b)$.
Substituting $f(0) \le f(b)$ into (2.8.4), $f(a) + f(b) \le f(a+b) + f(0) \le f(a+b) + f(b)$, and cancelling $f(b)$ from both sides gives

$$f(a) \;\le\; f(a+b),$$

which proves f is monotonically increasing for positive arguments. ■

Lemma 2.8.5: Convex Even Functions Are Subadditive

Lemma 2.8.5 (Convex even functions are subadditive)

Let f be an even convex function such that $f(0) \leq 0$. Then

$$f(a) + f(b) \ \leq \ f(a+b) \qquad \text{for all } a, b \geq 0.$$

Plain English. "Subadditive" means applying f to a sum is at least as large as summing f applied separately: the function doesn't "lose value" when you combine its arguments. This follows immediately from (2.8.4) once we additionally know $f(0) \leq 0$: then $f(a) + f(b) \leq f(a+b) + f(0) \leq f(a+b)$. ■

This innocuous-looking lemma is the key technical ingredient for the main result of this subsection: a single inequality that, depending on which convex even function f we plug in, yields a bound for several different distances at once.

Theorem 2.8.6: General Entropy Inequality

Theorem 2.8.6 (General entropy inequality) [Hut05b, Sec. 3.9.2]

Let $(y_1, \ldots, y_d)$ and $(z_1, \ldots, z_d)$ be elements of a d-dimensional (semi)probability simplex, i.e. $y_i \geq 0$ with $\sum_i y_i = 1$ (and $z_i \geq 0$ with $\sum_i z_i \leq 1$). Then for any function $f : \mathbb{R} \to \mathbb{R}$ that is convex, even, and satisfies $f(0) \leq 0$, we have

$$\frac{1}{2} \sum_{i=1}^{d} f(y_i - z_i) \; \leq \; f\left(\sqrt{\frac{1}{2} \sum_{i=1}^{d} y_i \ln \frac{y_i}{z_i}} \right).$$

Plain English. y_i and z_i are the probabilities that two distributions Y and Z assign to outcome i, out of d possible outcomes. The right-hand inner expression $\sum_i y_i \ln(y_i/z_i)$ is the KL divergence between Y and Z. The theorem says: for *any* convex, even "distance-shape" function f (with $f(0) \leq 0$), the total per-outcome discrepancy $f(y_i - z_i)$, summed and halved, is bounded by f applied to the square root of half the KL divergence. Plugging in different choices of f below gives several concrete, familiar distance bounds "for free".

Proof. Let $I = \{1, \dots, d\}$. Partition $I = P \dot\cup Q$ into two disjoint sets. For $S \in \{P, Q\}$, define $y^S := \sum_{i \in S} y_i$ and $z^S := \sum_{i \in S} z_i$, and normalize $p_i^S := y_i / y^S$, $q_i^S := z_i / z^S$ for $i \in S$; these are valid probability distributions on S.
By Gibbs' inequality (Theorem 2.5.15, which says KL divergence is always ≥ 0),

$$0 \;\leq\; \sum_{i \in S} p_i^S \ln \frac{p_i^S}{q_i^S} \;=\; \frac{1}{y^S} \sum_{i \in S} y_i \ln \frac{y_i}{z_i} \;-\; \frac{1}{y^S} \sum_{i \in S} y_i \ln \frac{y^S}{z^S},$$

hence $\sum_{i \in S} y_i \ln \frac{y_i}{z_i} \geq y^S \ln \frac{y^S}{z^S}$.

Proof of Theorem 2.8.6 II

Noting $y^Q = 1 - y^P$ and $z^Q \leq 1 - z^P$, and summing the P and Q contributions,

$$\sum_{i=1}^{d} y_i \ln \frac{y_i}{z_i} \;\geq\; y^P \ln \frac{y^P}{z^P} + y^Q \ln \frac{y^Q}{z^Q} \;\geq\; y^S \ln \frac{y^S}{z^S} + (1 - y^S) \ln \frac{1 - y^S}{1 - z^S} \;\geq\; 2(y^S - z^S)^2,$$

where the last step is exactly Pinsker's binary inequality (Lemma 2.8.1) applied with $p = y^S, q = z^S$.
Therefore

$$(y^S - z^S)^2 \;\leq\; \frac{1}{2} \sum_{i=1}^{d} y_i \ln \frac{y_i}{z_i}. \tag{2.8.7}$$

Proof of Theorem 2.8.6 III

Now choose $P = \{i \in I : y_i > z_i\}$ and $Q = \{i \in I : y_i \leq z_i\}$, and bound $\sum_{i \in S} f(y_i - z_i)$:

$$\sum_{i \in S} f(y_i - z_i) \overset{(a)}{=} \sum_{i \in S} f(|y_i - z_i|) \overset{(b)}{\leq} f\Big(\sum_{i \in S} |y_i - z_i|\Big) \overset{(c)}{=} f\Big(\Big|\sum_{i \in S} y_i - z_i\Big|\Big) \overset{(d)}{=} f(|y^S - z^S|)$$

$$= f\big(\sqrt{(y^S - z^S)^2}\big) \overset{(e)}{\leq} f\left(\sqrt{\tfrac{1}{2}\sum_{i=1}^{d} y_i \ln \frac{y_i}{z_i}}\right) .$$

Plain English of steps. (a) f is even, so $f(x) = f(|x|)$. (b) Lemma 2.8.5 (subadditivity of f). (c) By construction of S, all the terms $y_i - z_i$ inside S share the same sign, so we may sum first, then take the absolute value. (d) Distribute the sum. (e) Apply (2.8.7) together with monotonicity of $\sqrt{\cdot}$ and of f (Lemma 2.8.2).

Finally, we conclude by combining the bound for P and Q:

$$\frac{1}{2}\sum_{i=1}^{d} f(y_i - z_i) \;=\; \frac{1}{2}\sum_{i\in P} f(y_i - z_i) + \frac{1}{2}\sum_{i\in Q} f(y_i - z_i) \;\le\; f\left(\sqrt{\frac{1}{2}\sum_{i=1}^{d} y_i \ln \frac{y_i}{z_i}}\right). \qquad \blacksquare$$

Corollary 2.8.8: Specific Entropy Inequalities

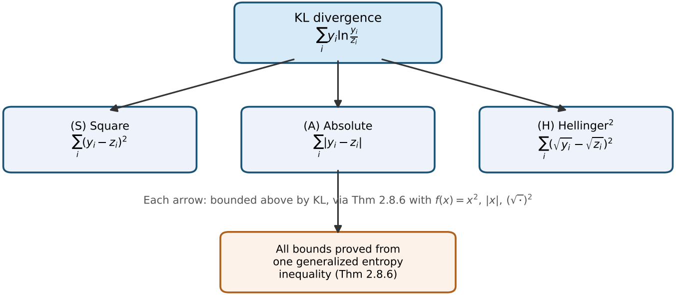
Corollary 2.8.8 (Specific entropy inequalities)

Continuing from Theorem 2.8.6:

$$
\begin{aligned}
\text{(S) Square:} \quad & \sum_{i=1}^{d} (y_i - z_i)^2 \;\le\; \sum_{i=1}^{d} y_i \ln \frac{y_i}{z_i} \\
\text{(A) Absolute:} \quad & \sum_{i=1}^{d} |y_i - z_i| \;\le\; \sqrt{2 \sum_{i=1}^{d} y_i \ln \frac{y_i}{z_i}} \\
\text{(H) Hellinger}^2: \quad & \sum_{i=1}^{d} (\sqrt{y_i} - \sqrt{z_i})^2 \;\le\; \sum_{i=1}^{d} y_i \ln \frac{y_i}{z_i}
\end{aligned}
$$

Plain English. Each row is one concrete, well-known distance between the distributions Y and Z, all bounded above by the very same KL divergence $\sum_i y_i \ln(y_i/z_i)$. Row (S) is obtained from Theorem 2.8.6 by choosing $f(x) = x^2$; row (A) by choosing $f(x) = |x|$; row (H) needs a slightly different, direct argument (below), since the Hellinger^2 distance is not of the simple $\sum f(y_i - z_i)$ form.

The Distance Hierarchy at a Glance



Proof of Corollary 2.8.8

Proof. Taking Theorem 2.8.6 and substituting $f(x) = x^2$ or $f(x) = |x|$ directly gives (S) or (A) respectively.
To prove (H), for $t, y, z > 0$ define

$$g(t) := -\ln(t) + t - 1 \;\geq\; 0$$

Plain English. $g(t)$ measures how far t is from 1 in a logarithmic sense; it is a standard fact that $g(t) \geq 0$ for all $t > 0$, with equality only at $t = 1$ (this is essentially Gibbs' inequality again).
This implies (after some algebra)

$$f(y, z) \;:=\; y \ln \frac{y}{z} - (\sqrt{y} - \sqrt{z})^2 + z - y \;=\; 2y\, g(\sqrt{z/y}) \;\geq\; 0,$$

hence $y \ln(y/z) - (\sqrt{y} - \sqrt{z})^2 \geq y - z$. Summing over $i = 1, \ldots, d$ (with $y = y_i$, $z = z_i$),

$$\sum_i y_i \ln \frac{y_i}{z_i} - \sum_i (\sqrt{y_i} - \sqrt{z_i})^2 \;\geq\; \sum_i y_i - \sum_i z_i \;\geq\; 1 - 1 \;=\; 0,$$

from which (H) follows (using $\sum_i z_i \leq 1$). ■

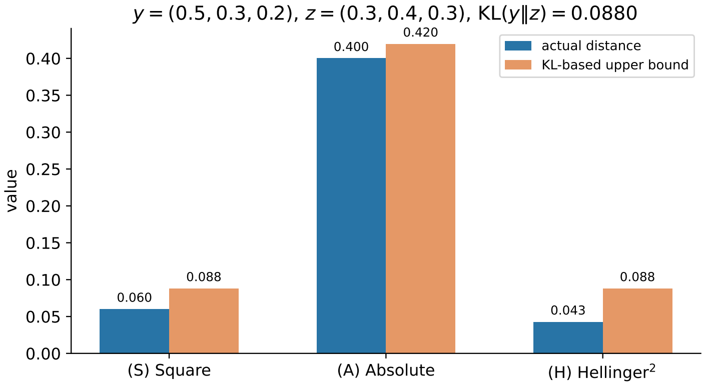
Python: Verifying the Entropy Inequalities Numerically

We numerically check Corollary 2.8.8 for a concrete pair of distributions $y = (0.5, 0.3, 0.2)$ and $z = (0.3, 0.4, 0.3)$ (both probability vectors on 3 outcomes).

```
import numpy as np

y = np.array([0.5, 0.3, 0.2])
z = np.array([0.3, 0.4, 0.3])

kl = np.sum(y * np.log(y / z))                          # KL(y || z)
square    = np.sum((y - z)**2)                           # (S)
absolute  = np.sum(np.abs(y - z))                        # (A)
hellinger = np.sum((np.sqrt(y) - np.sqrt(z))**2)         # (H)

print(f"KL={kl:.4f}  square={square:.4f}  absolute={absolute:.4f}  hellinger2={hellinger:.4f}")
# KL=0.0880  square=0.0600  absolute=0.4000  hellinger2=0.0427
```

Reading the output. All three bounds hold ($0.0600 \leq 0.0880$, $0.4000 \leq \sqrt{2 \cdot 0.0880} = 0.4196$, $0.0427 \leq 0.0880$), exactly as Corollary 2.8.8 guarantees.

Entropy Inequalities: Actual Distance vs. KL-Based Bound



Why Semimeasures Are a Headache

Recall (Definition 2.2.11) that a *semimeasure* ν relaxes a probability measure by allowing $\sum_a \nu(\Gamma_{xa}) \leq \nu(\Gamma_x)$ rather than requiring equality: some probability mass can simply be "missing".

The problem. Measure theory itself is already a mature, well-developed field. Semimeasure theory, by contrast, essentially does not exist as a developed field — yet semimeasures are exactly what Solomonoff induction and the AIXI agent are built on (via lower semicomputable semimeasures, class $\mathcal{M}_{sol}$). This subsection lists several pragmatic workarounds used throughout the book, since a full semimeasure theory is beyond its scope.

Only use ν on cylinder sets

For most of the book we only ever need to know ν on *cylinder sets* Γ_x (the set of all infinite sequences starting with the finite string x). The most frequently used quantity is the predictive distribution

$$\nu(x_t \mid x_{<t}) \equiv \nu(\Gamma_{x_{1:t}})/\nu(\Gamma_{x_{<t}}).$$

Plain English. This is simply "the probability of the next symbol given everything seen so far", written as a ratio of two cylinder-set masses. For predictions over a fixed finite lifetime m, the sample space is finite and this trivializes semimeasure theory entirely.

Developing or Deriving Semimeasure Results

Develop semimeasure theory

Extending measure theory's definitions and results to semimeasures could be an interesting research program, but is beyond the book's scope (left as an exercise for "eager math students").

Derive required semimeasure results

A more focused alternative: only develop what is actually needed for this book. Even this is a major undertaking. Extending *finitary* results (e.g. the entropy inequalities of Corollary 2.8.8) is often not too hard and has been done. For others — such as martingale convergence, or even the general definition of expectation — one would need to dig much deeper.

Defining ν on measurable sets

Strictly, ν is only a *pre*-semimeasure. To extend it to a full σ-algebra $\mathcal{F}$ we would need a semimeasure version of Carathéodory's extension theorem. The book sketches (but does not fully verify) a candidate construction using an *inner* semimeasure $\nu_*(A) := \sup_{\mathcal{S}:\Gamma_{\mathcal{S}}\subseteq A} \sum_{x\in\mathcal{S}} \nu(\Gamma_x)$ and an *outer* semimeasure $\nu^*(A) := \inf_{\mathcal{S}\supseteq A} \sum_{x\in\mathcal{S}'} \nu(\Gamma_x)$, noting this remains a genuinely open research problem.

Expectations With Respect to Semimeasures

Once we can evaluate ν on measurable sets, we could (mis)use the general definition of expectation for non-negative f:

$$\mathbb{E}_\nu[f] \;:=\; \int_0^\infty f^*(t)\,dt \qquad \text{with} \qquad f^*(s) := \nu(\{\omega \in \Omega : f(\omega) > s\}).$$

Plain English. This is the standard "layer-cake" formula for expectation: integrate, over all thresholds s, the probability that f exceeds s.
The catch: linearity breaks. Decomposing the constant function $1 = \sum_{a\in\mathcal{X}} f_a$ with $f_a(x_{1:\infty}) := [\![x_1 = a]\!]$ (an indicator that the first symbol is a), we get

$$\mathbb{E}\Big[\sum_a f_a\Big] \;=\; \mathbb{E}_\nu[1] \;=\; \nu(\Gamma_\epsilon) \;\geq\; \sum_{a\in\mathcal{X}} \nu(\Gamma_a) \;=\; \sum_a \mathbb{E}[[\![x_1 = a]\!]] \;=\; \sum_a \mathbb{E}[f_a].$$

Plain English. Γ_ϵ is the cylinder set for the empty string (the whole sample space). Because ν can leak mass, the expectation of a sum of indicators can strictly exceed the sum of their individual expectations — i.e. **expectation is no longer linear** once ν is only a semimeasure. Other definitions of $\mathbb{E}$ may preserve linearity, at the cost of other properties.

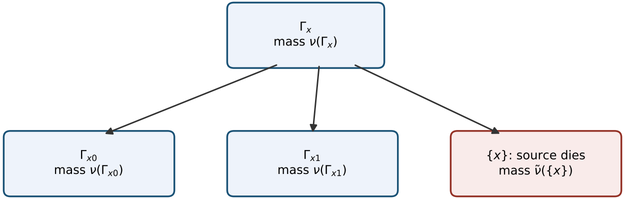
Include Finite Sequences: Modeling “Death”

Some monotone Turing machines may not output infinite sequences, but stop or loop forever after only a finite output. The most honest fix: enlarge the sample space from $\mathcal{X}^\infty$ to $\mathcal{X}^\infty \cup \mathcal{X}^*$, and convert any semimeasure ν into a genuine measure $\tilde{\nu}$ via

$$\tilde{\nu}(\Gamma_x^*) := \nu(\Gamma_x) \qquad \text{and} \qquad \tilde{\nu}(\{x\}) := \nu(\Gamma_x) - \sum_{a \in \mathcal{X}} \nu(\Gamma_{xa}).$$

Plain English. Γ_x^* is the cylinder set extended to also include the finite string x itself. $\tilde{\nu}(\{x\})$ — the “missing” probability mass at x — is interpreted as the probability that the source (the world, or the predictor/agent) *dies* exactly after emitting x. This is used later in Section 15.7. Because $\Gamma_x^* = \{x\} \cup \bigcup_a \Gamma_{xa}^*$, indeed $\tilde{\nu}(\Gamma_x^*) = \tilde{\nu}(\{x\}) + \sum_a \tilde{\nu}(\Gamma_{xa}^*)$ — so $\tilde{\nu}$ is a genuinely proper (non-leaky) probability measure.

Where the Missing Mass Goes



$\nu(\Gamma_x) \ = \ \nu(\Gamma_{x0}) + \nu(\Gamma_{x1}) \ + \ $(missing mass)
Extending the alphabet with a death symbol (or extending the sample space)
turns this missing mass into an honest probability $\tilde{\nu}(\{x\})$ of "the world ending".

Extending the Alphabet With a Death Symbol

A similarly "honest" idea: extend the alphabet to $\mathcal{X} \cup \{\perp\}$, adding a special "death" symbol $\perp$ (pronounced "bottom"), and extend μ via

$$\nu(\Gamma_{x\perp^k}) := \nu(\Gamma_x) - \sum_{a \in \mathcal{X}} \nu(\Gamma_{xa}) \quad \forall k \in \mathbb{N}^+, \qquad \nu(\Gamma_{x\perp y}) := 0 \quad \forall y \in (\mathcal{X} \cup \{\perp\})^* \setminus \{\perp\}^*.$$

Plain English. Once $\perp$ has occurred, it just keeps recurring forever (the world has "ended"); no other symbol can follow it. This keeps the convenience of working with proper infinite sequences (no need for the hybrid finite/infinite sample space above), while ν becomes a genuine measure on $(\mathcal{X} \cup \{\perp\})^\infty$.
Consequence for agents. If we assign reward 0 once $\perp$ is observed, the Bellman equations (used to define optimal value functions) remain valid, and agent death is faithfully modeled. One caveat: some results have only been derived for binary $\mathcal{X}$, so are not available once $\perp$ is added to a binary alphabet (making it effectively ternary).

Solomonoff Normalization

Solomonoff normalization

Given a semimeasure ν on cylinder sets, define

$$\nu_{norm}(x_t \mid x_{<t}) := \frac{\nu(\Gamma_{x_{1:t}})}{\sum_{x_t \in \mathcal{X}} \nu(\Gamma_{x_{1:t}})}, \quad \nu_{norm}(\Gamma_{x_{1:n}}) := \prod_{t=1}^{n} \nu_{norm}(x_t \mid x_{<t}), \quad \nu_{norm}(\epsilon) := 1.$$

Plain English. At every step, we simply rescale (renormalize) the semimeasure's predictive probabilities so they sum to exactly 1, then multiply these normalized one-step probabilities together. This gives a properly normalized probability measure that still *dominates* ν (i.e. $\nu_{norm}(\Gamma_x) \geq \nu(\Gamma_x)$), which is the key property needed later (Proposition 3.1.4). Downsides: ν_{norm} is no longer lower semicomputable (only limit-computable), even if ν was, and we lose the "semimeasure defect" that used to be attributed to death of an agent.

Lower-Limit Normalization

Lower limit normalization

We could instead look for the *largest measure* $\underline{\nu} \leq \nu$. This is unique, and can be represented [Hut14b] as

$$\underline{\nu}(\Gamma_{x_{<t}}) := \lim_{n\to\infty} \sum_{x_{t:n}} \nu(\Gamma_{x_{1:n}}).$$

Plain English. We push all remaining probability mass as far into the future as possible before "giving up" on it, and take the limit. The limit exists because the sum is decreasing in n (a consequence of ν being a semimeasure): $\underline{\nu}$ is the "most pessimistic" proper measure consistent with never assigning more mass than ν itself does.

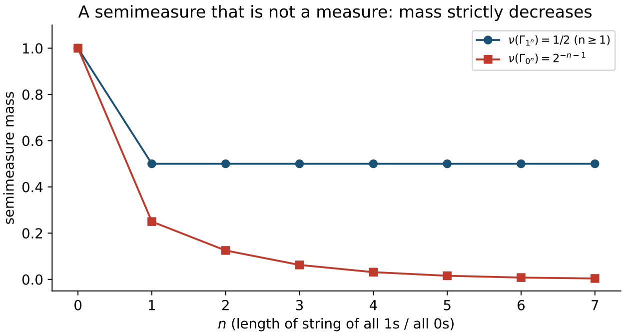
Python: A Concrete Semimeasure That Is Not a Measure

The book gives a small concrete example (p. 114): a semimeasure ν with $\nu(\Gamma_{1^n}) = 1/2$ for all $n \geq 1$, $\nu(\Gamma_{0^n}) = 2^{-n-1}$, and $\nu = 0$ on every other sequence prefix.

```python
import numpy as np

n_max = 6
nu_1n = np.array([0.5] * n_max)                          # nu(Gamma_{1^n}), n=1..n_max
nu_0n = np.array([2.0**(-n-1) for n in range(1, n_max+1)])  # nu(Gamma_{0^n})

print("nu(Gamma_{1^n}):", nu_1n)
print("nu(Gamma_{0^n}):", nu_0n)

# nu(Gamma_0) = 1/4 directly, but the LARGEST MEASURE below nu (underline_nu)
# assigns underline_nu(0) = lim_{n->inf} nu(Gamma_{0^n}) = 0
print("nu(0)=0.25  underline_nu(0)=0.0  underline_nu(eps)=0.5")
```

Plain English. Even though ν assigns positive mass $1/4$ to "starts with 0", that mass keeps shrinking geometrically the longer the run of 0s must continue, all the way down to 0 in the limit. So the lower-limit-normalized measure $\underline{\nu}$ (which looks arbitrarily far into the future) assigns 0 to that same cylinder set — illustrating $\underline{\nu} \neq \nu$ in general, and that $\underline{\nu}$ is not itself a probability measure ($\underline{\nu}(\Gamma_\epsilon) = 1/2 \neq 1$) until renormalized.

A Semimeasure That Is Not a Measure: The Plot



Restricted Classes of Measures I

Computable measures $\mathcal{M}_{comp}$. A monotone Turing machine that is *guaranteed* to output only infinite sequences generates a properly normalized probability measure. Let $\mathcal{M}_{comp}$ be the class of all such measures. Each measure in $\mathcal{M}_{comp}$ is computable, but — unlike $\mathcal{M}_{sol}$, the class of lower semicomputable semimeasures — $\mathcal{M}_{comp}$ is not effectively enumerable, hence not approximable by an algorithm that lists it out.

Global assumption: μ is a proper probability measure. This is assumed throughout the rest of the book. Some key theorems (such as the entropy inequalities, Corollary 2.8.8) genuinely fail if the true environment μ is only a semimeasure. There is generally no pressing need to model semimeasure environments directly, except possibly to model finite universes or death.

Primitive recursive measures $\mathcal{M}_{pr}$. Instead of considering all Turing machines, restrict to *primitive recursive* (p.r.) functions — total functions with a primitive-recursive upper bound on their own run-time. This class contains virtually every computable function one usually encounters in practice; functions that are computable but not p.r. are extremely slow-growing (e.g. functions growing like the Ackermann function). The class of p.r. functions is itself recursively enumerable [Liu60], so it can be converted into an enumeration of p.r. measures $\mathcal{M}_{pr} \subset \mathcal{M}_{comp}$, e.g. via Solomonoff normalization. Assuming a computable prior weight w_ν, the Bayesian mixture ξ over $\mathcal{M}_{pr}$ is then computable — but, perhaps surprisingly, $\xi \notin \mathcal{M}_{pr}$ itself (mixing takes you outside the class you started with).
Reflective-oracle computable measures $\mathcal{M}_r^O$. In Section 10.5 we will meet a strictly larger class $\mathcal{M}_r^O \supset \mathcal{M}_{comp}$ of measures computable with the help of a *reflective oracle*. It has the remarkable property that the normalized Bayes mixture both dominates the class *and* is itself a member of the class; even the Bayes-optimal policy π_ξ^* is in the class, solving the long-standing *Grain of Truth problem* (the question of whether an agent's own prior belief class can contain the true environment together with rational agents reasoning about each other).

Conclusion of Section 2.8.2

Ideally, the *entire* book would be developed for semimeasures, since $\mathcal{M}_{sol}$ (lower semicomputable semimeasures) is the core class of interest for Solomonoff induction and the AIXI agent. But this is unfortunately not feasible with current mathematical tools. As a consequence, most of the theory in this book is developed for proper measures only, and the workarounds above are applied whenever semimeasures unavoidably crop up.

Rule of thumb, if in doubt. Assume the death symbol $\perp$ has been added to the alphabet $\mathcal{X}$ (and, in the agent case, assume reward $r = 0$ once $\perp$ is perceived), or simply restrict attention to $\mathcal{M}_{pr}$ or $\mathcal{M}_{comp}$ instead of the full $\mathcal{M}_{sol}$.

Why Probability-Zero Events Matter

For continuous sample spaces Ω, probability-zero events are everywhere. Example: throwing darts at a dartboard — the probability of hitting any single, specific point is exactly zero, even though every player's dart lands *somewhere*, so some probability-zero outcome always occurs. Technically this is modeled using a probability *density* rather than a probability mass. Conditional probabilities and expectations conditioned on probability-zero outcomes can still be defined, but require care.

Good news for this book. Nearly all of the book deals only with *discrete* random variables, which simplifies matters considerably — but does not entirely trivialize the problem, as we will see. This subsection collects the relevant issues and the (single) convention adopted throughout the rest of the book, so as not to clutter later chapters with repeated caveats.

Expectation for Partial Functions

Recall the definition of expectation (Definition 2.2.36) works cleanly as long as the function g is *total*. If g can take infinite values, or is only *partial* (undefined on some inputs), the correct definition for discrete X is

$$\mathbb{E}[g(X)] = \sum_{x \in X(\Omega): p(x) > 0} g(x)\, p(x) \qquad \text{for } g : \mathbb{R} \to \mathbb{R} \cup \{\pm\infty\} \cup \{\perp\}.$$

Plain English. We simply skip any outcome x that has zero probability $p(x) = 0$ before summing — so it never matters whether $g(x)$ is well-defined there. For total, finite-valued g this restriction is redundant and can be dropped, but we still want $\mathbb{E}$ to be defined even when g is infinite or undefined ($\perp$) at some zero-probability point.
Worked example. Let $\Omega = \{0, 1\}$ and $g(x) := 1/p(x)$. If $\forall x.\, p(x) > 0$, then $\mathbb{E}[g(X)] = \sum_{x \in \Omega} 1 = 2$. But if $p(0) = 0$ (so $g(0)$ is undefined, dividing by zero), we simply get $\mathbb{E}[g(X)] = g(1)p(1) = 1$: the problematic point $x = 0$ is silently excluded from the sum because it has probability zero.

Conditioning on Probability-Zero Events

Conditioning on $P(A) = 0$

Since a probability-zero event will (almost) never occur, the *safest* option is to simply leave $P(B \mid A)$ **undefined** whenever $P(A) = 0$. But this is restrictive (it does not allow conditional *densities*, needed in continuous settings) and cumbersome (it requires special-casing every theorem). There is no fully satisfactory general solution to this problem; see [Hut05b, Sec. 3.9.1] for some unorthodox suggestions.

Conditioning on $\mu(x_{<t}) = 0$: the convention used here

We only encounter this issue for random sequences and $\mu(\cdot \mid x_{<t})$ when $\mu(x_{<t}) = 0$, so we focus on this one special case. **The adopted convention:** add the qualifier "holds with μ-probability 1" to every theorem statement, if it is not there already. Since $\mu(x_{<t}) = 0$ is itself a probability-zero event, we are then free to *safely ignore* such histories $x_{<t}$ altogether when checking whether a theorem's conclusion holds.

Probability Kernels

In the agent setting, we consider a policy π interacting with an environment μ, generating a history $h_{1:\infty}$. We sometimes need to know how a policy π *would* continue a history $h_{<t}$ even if $h_{<t}$ itself has probability 0 under π (e.g. because $h_{<t}$ was actually generated by a *different* policy).

The solution: probability kernels. Start with *conditional* distributions directly, and take their product. Formally, a *probability kernel* is a family of functions $\mu_t : \mathcal{X}^t \to \Delta\mathcal{X}$ for each $t \in \mathbb{N}$, where for every history $x_{<t}$, $\mu_t(\cdot\,; x_{<t})$ is a genuine probability distribution over the next symbol $\mathcal{X}$ (note the semicolon — it is not the same as a conditional probability derived *from* a joint measure).

$$\mu(x_{1:n}) := \prod_{t=1}^{n} \mu_t(x_t; x_{<t}) \qquad \text{and} \qquad \mu(x_t \mid x_{<t}) := \mu_t(x_t; x_{<t}).$$

Plain English. This product defines a genuine joint measure μ, and the conditional $\mu(x_t \mid x_{<t})$ is *uniquely and well-defined even when* $\mu(x_{<t}) = 0$ — because we started from the conditionals themselves rather than trying to derive them via division. The collection of kernels (μ_t) therefore carries strictly *more* information than the joint measure μ alone.

Probabilistic Turing Machines and Kernels

Consider a probabilistic Turing machine T that lower semicomputes semiprobability kernels, i.e. $\nu_T(x_t \mid x_{<t})$ is the probability that $T(x_{<t}, t, \text{random noise})$ outputs x_t.
A subtly smaller class. The class $\{\nu_T : T \text{ a TM}\}$ is recursively enumerable, and contains *all and only* the semimeasures that are *conditionally* lower semicomputable. This class is "slightly" smaller than $\mathcal{M}_{sol}$, since a semimeasure $\nu \in \mathcal{M}_{sol}$ is not automatically *conditionally* lower semicomputable.

Pragmatic approach (used throughout the book)

Take the approach above, but without being pedantic about the semicolon notation and the time index t: simply write $\nu(x_t \mid x_{<t})$ directly, even though strictly speaking it is a probability kernel and not literally a conditional distribution when $\nu(x_{<t}) = 0$.

Worked example. If $\nu_T^s(x_{<t})$ (for $s = 1, 2, 3, \ldots$) is a lower-computation sequence approximating $\nu_T(x_{<t})$ from below, define $\nu_T^s(x_t \mid x_{<t}) := 1/|\mathcal{X}|$ (or any other computable placeholder) as long as $\nu_T^s(x_{<t}) = 0$. Once (if ever) $\nu_T^s(x_{<t}) > 0$, switch to $\nu_T^s(x_t \mid x_{<t}) := \nu_T^s(x_{1:t})/\nu_T^s(x_{<t})$. Note the resulting conditional distribution is generally *not* lower semicomputable, but only *limit-computable* in every case.

2.8.4 Exercises (Section 2.8)

The book poses seven exercises testing the material of Section 2.8:

1. Prove the Absolute distance and the Un-Squared Hellinger distance $\sqrt{H}$ are both metrics, but the KL divergence is not.
2. Show the bounds in Corollary 2.8.8 are tight: the ratio of left-hand side to right-hand side can get arbitrarily close to 1.
3. Generalize all distance definitions and bounds to probability measures on general sample spaces Ω, except for the 2-norm; explain what goes wrong for the 2-norm.
4. Derive an explicit expression for $\tilde{\nu}(\Gamma_x)$ (note: $\Gamma_x \neq \Gamma_x^*$) and show $\underline{\nu}(\Gamma_{x<t}) = \tilde{\nu}(\Gamma_{x<t})$, both defined in Section 2.8.2.
5. Prove all claims made in Section 2.8.2.

6–7. (Open-ended, research-level) Suitably generalize all results in this book to semimeasures, or extend a measure-theory textbook's definitions and results to semimeasures, and prove them.

Purpose of This Section

Section 2.9 is the book's guide to the wider literature behind every topic covered in Chapter 2. It is organized to mirror the chapter's own subsections (2.1 through 2.8), giving, for each, the key historical breakthroughs and the standard textbooks/papers a reader could turn to for more depth.

Two anchor references for AI in general.

- *Artificial Intelligence: A Modern Approach* [RN20] — the canonical AI textbook, comprehensive overview with over a thousand further AI-related references.
- *An Introduction to Kolmogorov Complexity and Its Applications* [LV19] — its introductory chapter covers similar background material, with many further related references.

This history section itself is an updated version of an earlier one from the author's PhD-thesis-turned-monograph [Hut05b, Sec. 2.5].

A Timeline of Selected Milestones

Binary strings (2.1). Based on the corresponding sections in [Hut05b] and [LV19]. *Principles of Mathematical Analysis* by Rudin [RT64] gives an introduction to the real-analysis concepts assumed throughout. Prefix codes are discussed in depth in Li and Vitányi's book [LV19]; the general theory of coding and prefix codes is covered in [Gal68].

Measure theory and probability (2.2). Based on material from [GS20]. *Real Analysis* by Stein and Shakarchi [SS09] gives a slower, more formal treatment of measure theory. The i.i.d. assumption is prevalent in statistics and machine learning, but unsuitable for AGI [hut22].

A brief history of probability. Games of chance date back to around 300 B.C., but the first *mathematical analysis* of probability appears much later. Key breakthroughs, chronologically: a systematic way of calculating probabilities by Cardano [Car63]; a systematic treatment via Pascal (in correspondence with Fermat) and conditional probability [Pas54]; Bayes' rule [Bay63]; the distinction between subjective and objective probability, and the weak law of large numbers, by Bernoulli [Ber13]; equi-probability due to symmetry by Laplace [Lap12]; the principle of indifference by Keynes [Key21]; Kolmogorov's axioms of probability theory [Kol33]; early attempts to define randomness of individual sequences by von Mises [Mis19], Wald [Wal37], and Church [Chu40], finally made successful by Martin-Löf [ML66]; and the notion of a universal a priori probability by Solomonoff [Sol64], mathematically investigated by Levin [ZL70a, Lev74].

Recommended Probability Textbooks and Statistical Inference (2.3)

Textbooks for probability theory. *Probability and Random Processes* by Grimmett et al. [GS20] is a gentle introduction with plenty of exercises (the basis for most of this chapter). *Introduction to Probability Models* by Ross [Ros14] is more advanced, with a similar structure but going well beyond this book's scope. For the early history of probability, see Hacking [Hac75] and Hald [Hal90]; for the 20th-century foundations, see Schnorr [Sch71] and the PhD theses of van Lambalgen [Lam87] and Wang [Wan96]. See also Shafer & Vovk [SV01] for a game-theoretic/finance perspective, and Feller [Fel68] as a standard textbook. A philosophical treatise on conditional probability is [Haj11].
Statistical inference and estimation (2.3). A great introduction to frequentist statistics, focusing on general theory over specific models, is Wasserman [Was10]. Section 2.3.4 is based on a chapter from *Elements of Information Theory* [CT06], where the Cramér–Rao inequality (Theorem 2.3.14) is proven; the "sunrise problem" was first described by Laplace [Lap40], where the rule of succession now known as *Laplace's Rule* was introduced (Section 2.4.3).

2.4 Bayesian Probability Theory I

The limitations of frequentist statistics are discussed in [Hut05b, Sec. 2.3.1][Háj09b]: the naive definition is *circular* (though see Cournot's forgotten principle [Cou43]); identical events are a mirage (the *reference class problem*); and the most serious limitation is the assumption of i.i.d. data.

Alternatives to Bayesian reasoning in AI's history. Default reasoning [Rei80]; non-monotonic logic [MD80]; circumscription [McC80]; certainty factors [Sho76, BS84]; Dempster–Shafer theory [Dem68, Sha76]; fuzzy logic [Zad65, Zim91]; possibility theory [Zad78]; imprecise probabilities [Wal91, Hut03f, ZH05, PZTH07, Hut09h, PZTH09]; robust Bayesian analysis [RIR00]; and many others (see [Fin73, Wal91, RN20] for a more detailed account).

Why classical probability theory fell briefly out of favor, and its return. Many reasons were put forward in the 1970s: strict numerical values seemed inappropriate for qualitative reasoning, probability theory cannot deal with impreciseness/vagueness/subjective belief, or is just impractical. But all these alternative approaches turned out to have *their own* problems — unclear semantics, lack of self-consistency, or failure to scale. Bayesian and frequentist probability theory leave the fewest open questions, especially Bayesianism, with solid justifications via Cox's axioms [Cox46] and Dutch book arguments [Háj09a, Pre09, Sec. 2]. The debates [Che85, Che88] have essentially settled in the new millennium in favor of machine-learning systems based on classical statistics [HTF09] and increasingly (sometimes robust) Bayesian methods [Bis06, KF10, Mur22, Mur23] — returning to its roots of 1763 [Bay63].

2.4 Bayesian Probability Theory III

Bayes' Law and Laplace's Rule. Bayes' Law is one of the earliest non-trivial rules in probability and statistics [Bay63], which *Laplace's Rule* [Lap40] builds upon. Jaynes' philosophical text [Jay03] treats probability theory as a natural extension of (Boolean) logical reasoning; Earman [Ear93] is another philosophical text on Bayes; Gelman [GCSR95] is more practical (data analysis); Berger [Ber93] sits in-between. More on Bayesian inference and choices of model class/prior: [BC16, GvdV17]; standard Bayesian machine-learning textbooks: [Bis06, Mur22, Mur23].

Objective vs. subjective probability. An ongoing debate, sharpened in the 20th century. Advocates of the relative-frequency / objective interpretation: Kolmogorov [Kol63], Fisher [Fis22], von Mises [Mis28]. Hajek [Háj96, Háj09b] offers fifteen arguments against finite/hypothetical frequentism. Advocates of probability as degrees of belief: Popper [Pop34], Ramsey [Ram31], Finetti [Fin37, Fin74], Cox [Cox46], Savage [Sav54], Jeffreys [Jef83], Good [Gol06]. See [OBD+06, O'H19] on eliciting subjective probabilities from experts; objective priors are advocated in [Ber06, BBS24], and "universal priors" in [Hut07e, RH11]. Jeffreys' objective prior (Definition 2.4.11) and its reparameterization invariance is discussed further in [Pre09, O'H10, Lee12]. See [Goo71] for a classification of "46656 varieties of Bayesians", and [Pre09, Sec. 2] for an overview of justifications of the Kolmogorov axioms from various perspectives.

Inductive logic and the reference class problem. Carnap [Car48, Car50] tried to supplement logic with probability theory, giving so-called *inductive logic*. This works for propositional logic [Jay03], but extending it satisfactorily to predicate logic failed for a long time [Put63], until finally solved [GS82, HLNU13b, HLNU13a, GBTC+20]. The closely related *reference class problem* is addressed in [Rei49, Kyb77, Kyb83, BGHK92].

2.5 Information Theory and Coding I

This section is based on the texts [Mac03, CT06, JJHH03, LV19]. The entire discipline of information theory originates from Shannon's original paper [Sha48], formalizing the problem of transmitting a message over a noisy communication medium, and introducing entropy while proving the source-coding theorem (Theorem 2.5.22).

Estimating and predicting entropy. The entropy of English text has been estimated through experiments using both humans [Sha51] and algorithms [Gue09, CK78] as predictors. The Kraft inequality is due to Kraft [Kra49]. The Kullback–Leibler divergence originates in a paper by the eponymous pair [KL51].

Coding methods. Shannon–Fano coding refers to two similar coding methods, by Shannon [Sha48] and Fano [Fan49]. Huffman coding [Huf52] is provably optimal given each symbol coded individually with a known frequency distribution. Better results come from block codes such as arithmetic codes [Pas76, HV91], which outperform Huffman coding on low-entropy messages. Golomb codes [GVV75] are optimal when symbols follow a geometric distribution. Pre-coding transformations like *run length encoding* [RC67] or *move-to-front* [Rya80], or the Burrows–Wheeler transform [BW94], can be applied before a further code to shrink the message more.

Dictionary and mixture-based compressors. Dictionary coders compress by replacing a substring with an index into a lookup table: the Lempel–Ziv family starts with LZ77 [ZL77] and LZ78 [ZL78], popularizing this technique and extending to LZW [Wel84], LZRW [Wil91b], and LZSS [SS82]. *Context mixing compressors* are similar to Context Tree Weighting (covered in Chapter 4): they estimate a probability distribution as a mixture of models and compress with respect to it using arithmetic coding [Sai04]. The PAQ family [Mah05] takes a weighted average of models; later improved using a neural network for the mixture [Mah07].

Surveys and the Hutter Prize. A survey of compression techniques, with trade-offs between compression time, compression ratio, and decompression time, is given in [Mah12, SM10]; further comparisons in [Mah22, Nem22]. The **Hutter Prize** [Hut20a] offers a monetary reward for compressing `enwik9` [Mah11], a 1GB sample of Wikipedia text, smaller than the (then-current) record of $\approx$ 113MB.

2.6 Computability Theory

Based on the texts [HMU06, Hut05b]. Hopcroft, Ullman and Motwani [HMU06] gives a very readable elementary introduction to automata theory, formal languages, and computability theory. [BBJ02, Sip12] also serve as good introductions; complexity theory (which additionally accounts for computation *time*) is covered by books ranging from easy [Sip12] to hard [AB09] to very advanced [HO02].

Founding results. Turing [Tur36] introduced the Turing machine and demonstrated the undecidability of the halting problem. Turing machines are formally equivalent to partial recursive functions (see [Rog67, Odi89, Odi99] for an introduction) as well as to Alonzo Church's lambda calculus [Chu36]. The halting problem corresponds to Gödel's incompleteness theorem [Göd31, Sho67], whose proof rests on a diagonal argument invented by Cantor [Can74, Dau90]. The works [Göd31, Kle36, Tur36, Pos44, ZL70a, Sch02a] together establish the importance of the various computability concepts of Section 2.6.3. The naming of estimable and approximable functions in the context of universal priors traces to [Hut03b, Hut06b]; the introduction to the arithmetic hierarchy follows [Nie09, Lei16b].

2.7 Kolmogorov Complexity I

Based on the corresponding sections of [LV19, Hut05b]; other introductory texts include [Cal02, Cha87, DH10]; see [Hut07a, Hut08b, SH15a] for short, light encyclopedic introductions.
A coarse early history of algorithmic information theory. Kolmogorov [Kol65] and Chaitin [Cha66] independently suggested defining the information content of an object as the length of the shortest program computing it [Hut08b]. Solomonoff [Sol64] independently invented the closely related universal prior probability distribution and used it for binary sequence prediction [Sol78, HLV07]. Levin worked out most of the mathematical details [ZL70a, Lev74], and invented the fastest algorithm for function inversion and optimization (up to a huge constant factor) [Lev73b, Gag07]. Chaitin's major focus [Cha66, Cha75] is on the halting probability $\Omega = \sum_{p:U(p)\ halts} 2^{-\ell(p)}$, the probability that a random program (weighted by length) causes the universal machine U to halt. See [Hut07a, LV07] for a short introduction and applications list.

Short compilers and description length principles. Assumption 2.7.8 (a short compiler) is an effective version of Kolmogorov's assumption that "reasonable" universal Turing machines yield coincidentally similar complexities [Kol65]. The Minimum Description Length (MDL) principle [Ris78, Gru07, GR19] and the Minimum Message Length (MML) principle [WB68, Wal05] are closely related information-theoretic approaches to model selection, both trading off model complexity against the size of the message needed to describe the data given the model. General consistency of MDL beyond i.i.d. data has been proven in [PH05a, Hut09b]; for an application to kernel learning, see [HHO23]. MDL with over-parameterized models is problematic [DSYW21]; the *prequential* MDL approach solves this in theory [DV99, PH05a] and in practice [BLH23, LBH23, RTBP+23]. MDL has been extended to reinforcement learning [Hut09d, Hut09e, Hut09f, Hut09a, Ngu13, Das16] and applied to control [MKSB22]. The Loss Rank Principle (LoRP) generalizes MDL to arbitrary loss functions and non-parametric models [Hut07d, TH10, HT10].

2.7 Kolmogorov Complexity III

Properties and variants of Kolmogorov complexity. The Invariance Theorem 2.7.6 is due to [Sol64, Kol65, Cha66]; Theorems 2.7.13 and 2.7.23 (symmetry of information, Theorem 2.7.18) to [Lev74, ZL70a, Kol83]; other (in)equalities are elementary. There are many variants of Kolmogorov complexity: the prefix complexity K used in this book [Lev74, Gác74, Cha75]; the earliest, "plain" Kolmogorov complexity C [Kol65]; process complexity [Sch73]; monotone complexity Km [Lev73a]; uniform complexity [Lov69b, Lov69a]; Solomonoff's universal prior $M = 2^{-KM}$ [Sol64, Sol78]; Chaitin's complexity Kc [Cha75]; extension semimeasure Mc [Cov74]; predictive complexity KP [VW98]; and others. They often differ from K only by $O(\log K)$, but otherwise share similar properties. For the relation between Kolmogorov complexity and Shannon's information theory, see [Kol65, Kol83, ZL70a, CT06]; [AM20] proves that average Kolmogorov complexity converges to Shannon entropy for stationary random sequences at the optimal rate.

2.7 Kolmogorov Complexity IV

Drawbacks and remedies; applications. The main drawback of these Kolmogorov complexity variants is that they are not finitely computable [Kol65, Sol64]. They can be approximated *from above* (Theorem 2.7.28), but no accuracy guarantee can be given; worse, the runtime needed for reasonable accuracy for $K(n)$ grows faster than any computable function of n. This led to *time-bounded* or *resource-bounded* complexity/probability that is finitely computable [Dal73, Dal77, FMG92, Ko86, PF97, Sch02b]. In particular, Levin complexity [Lev73b] adjusts K by adding a logarithmic penalty on the program's running time. Some probabilistic approximations of K: [Con97] (evolving short Lisp programs), [BMdR+14] (probabilistic approximation).
Li's *normalized compression distance* (NCD) [Li03] (building on Bennett [Ben98]) is a *universal similarity metric* over strings, measuring how much one string helps describe another; NCD minorizes every other computable distance measure. It is itself incomputable, but computable approximations exist (replacing K with the best available data compressor). Applications: clustering [CVW04, CV05], phylogenetic trees [HRR06, RA11], file-type classification for computer forensics [Axe10], virus analysis (biological [PP17, MRNA20, CV22] and digital [Bor16, GB03]), a metric on multisets [CV14], and (replacing K with Google hit counts) a metric on websites [CV07].

Miscellaneous historical remarks. Chaitin [Cha91] speculated on the computational power of the evolutionary information-gathering process and its relation to algorithmic information. Schmidt [Sch99] argued that (time-bounded) Kolmogorov complexity helps, rather than prevents, the search for extraterrestrial intelligence (SETI). Vovk [VW98] described universal portfolio selection schemes. Hutter [Hut10a] argues that a Theory of Everything needs to account for the Kolmogorov complexity of the location of the observer in the universe. A more personal account of the past, present, and future of algorithmic randomness as a foundation of induction and AI, for a general audience, can be found in [Hut11].

A discussion of Pinsker's inequality and the related **Bretagnolle–Huber inequality** can be found in [Can22]. That same reference also discusses:

- **Gibbs' variational principle** — a variational characterization of the KL divergence used throughout information theory and statistical mechanics;
- the **Donsker–Varadhan** representation — another variational formula for the KL divergence, important in large deviations theory;
- f-**divergences** — the broad family of divergences (of which KL, squared Hellinger, and total variation are all special cases) obtained by choosing different convex functions f, directly generalizing the construction behind Theorem 2.8.6 and Corollary 2.8.8 of this chapter.

The specific proof of Pinsker's binary inequality (Lemma 2.8.1) presented earlier in this deck has been taken from Wu's lecture notes [Wu17].

Chapter 2 Wrap-Up

What Sections 2.8–2.9 add to the chapter

- A toolbox of **distance bounds**, all routed through KL divergence, that will be reused constantly from Chapter 3 onward.
- A candid, technical discussion of why **semimeasures** are hard, together with the concrete workarounds (finite sequences, death symbols, Solomonoff/lower-limit normalization, restricted classes $\mathcal{M}_{pr}$, $\mathcal{M}_{comp}$, $\mathcal{M}_r^O$) used elsewhere in the book.
- A principled convention for handling **probability-zero** conditioning, via probability kernels and the "holds with μ-probability 1" qualifier.
- A comprehensive map of the **literature** underlying every topic in Chapter 2 — probability, Bayesian inference, information theory, computability, and Kolmogorov complexity — tracing each back to its historical origin.

This concludes Chapter 2 (Background). Chapter 3 begins to build the theory of **Universal Prediction** on top of these foundations.